

51.504 Machine Learning (2025)

Homework 3

Due 9 Dec 2025, Tuesday, 11.59pm

Question 1

- (a) [10 points] What is the output when we convolve the left image with the 3×3 filter on the right (assuming stride = 1 and no padding)?

$$\begin{array}{|c|c|c|c|c|} \hline 1 & 1 & 2 & 0 & 3 \\ \hline 2 & 4 & 2 & 1 & 1 \\ \hline 1 & 1 & 5 & 3 & 0 \\ \hline 1 & 2 & 6 & 0 & 1 \\ \hline 0 & 1 & 2 & 0 & 3 \\ \hline \end{array} \times \begin{array}{|c|c|c|} \hline 1 & 0 & -1 \\ \hline 0 & 1 & 0 \\ \hline -1 & 0 & 1 \\ \hline \end{array} = \begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline & & \\ \hline \end{array}$$

- (b) [10 points] What is the output if we instead perform convolution using padding = 1, stride = 2, and then apply ReLU?

Question 2

Let $X = (X_1, X_2, X_3, X_4)$ be the random variable with $0 \leq X_i \leq 4$ for $i = 1, 2, 3, 4$, and density function

$$f_X(x) = \frac{1}{Z_1} \exp\{x_1 x_2 x_3 x_4\}, \quad Z_1 = \int_0^4 \int_0^4 \int_0^4 \int_0^4 \exp\{x_1 x_2 x_3 x_4\}.$$

Let $Y = (Y_1, Y_2, Y_3, Y_4)$ be the random variable with $0 \leq Y_i \leq 4$ for $i = 1, 2, 3, 4$, and density function

$$f_Y(y) = \frac{1}{Z_2} \exp\{y_1 y_2 y_3 + y_1 y_2 + y_4\}, \quad Z_2 = \int_0^4 \int_0^4 \int_0^4 \int_0^4 \exp\{y_1 y_2 y_3 + y_1 y_2 + y_4\}.$$

Let G be the graph in Figure 1.

- (a) [5 points] Show that G is a Markov random field for Y but not X .

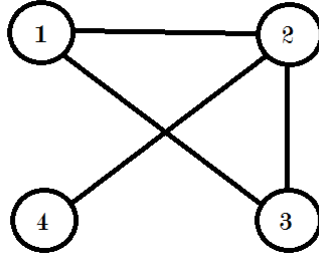


Figure 1: Graph G

- (b) [5 points] Write down a conditional independence relationship for Y that can be read off the graph G due to the global Markov property. (Note that Y_i is the random variable associated with node i .)
- (c) [5 points] Write down the conditional independence relationship for Y that follows from the local Markov property for node 4.
- (d) [5 points] Draw the appropriate arrows on G for the Bayesian network associated with the following set of structural equations:

$$\begin{aligned}
 Y_1 &= -2Y_3 + \epsilon_1, \\
 Y_2 &= 4Y_1 + Y_3 + \epsilon_2, \\
 Y_3 &= \epsilon_3, \\
 Y_4 &= -3Y_2 + \epsilon_4, \\
 \epsilon_i &\sim \mathcal{N}(0, 1).
 \end{aligned}$$

Question 3

Consider a mixture of two Gaussians, $\mathcal{N}(\mu_1, \sigma_1^2)$ with density function $f_1(x)$ and proportion π_1 , and $\mathcal{N}(\mu_2, \sigma_2^2)$ with density function $f_2(x)$ and proportion π_2 . The EM algorithm iteratively updates the parameters with the E-step,

$$\gamma(z_k^{(i)}) = \frac{\pi_k C_k \exp\{-(x^{(i)} - \mu_k)^2 / 2\sigma_k^2\}}{\sum_{j=1}^2 \pi_j C_j \exp\{-(x^{(i)} - \mu_j)^2 / 2\sigma_j^2\}},$$

and the M-step,

$$\begin{aligned}\pi_k &= \frac{1}{2} \sum_{i=1}^2 \gamma(z_k^{(i)}), \\ \mu_k &= \frac{\sum_{i=1}^2 x^{(i)} \gamma(z_k^{(i)})}{\sum_{i=1}^2 \gamma(z_k^{(i)})}, \\ \sigma_k^2 &= \frac{\sum_{i=1}^2 (x^{(i)} - \mu_k)^2 \gamma(z_k^{(i)})}{\sum_{i=1}^2 \gamma(z_k^{(i)})}.\end{aligned}$$

Suppose we perform EM on the sample data set $\{1, 1.5\}$. Let our initial parameter estimates be $\mu_1 = 0, \mu_2 = 3, \sigma_1 = 1, \sigma_2 = 2$. Also let our initial mixture parameters be $\pi_1 = 1/3$. We then have $f_1(1) = 0.241, f_1(1.5) = 0.13, f_2(1) = 0.12$ and $f_2(1.5) = 0.15$.

- (a) [10 points] Calculate the probabilities that each data point belongs to each Gaussian given the initial parameters.
- (b) [10 points] Compute the next iteration of $\pi_1, \pi_2, \mu_1, \mu_2$.
- (c) [10 points] Compute the next iteration of σ_1^2 and σ_2^2

Question 4

Consider the irregular neural network N in Figure 2. Let i_0, i_1 and i_2 be the input nodes and o_0, o_1, o_2 be the output nodes. Let x_0, x_1 and x_2 be the input variables of i_0, i_1 and i_2 respectively. Let the activation function at nodes h_0, h_1, o_0, o_1 and o_2 be $\sigma(x)$.

- (a) [10 points] Write down the output of o_2 in terms of x, w and σ .
- (b) [5 points] Which nodes' error signals δ contains the weight w_1 ?

For the remainder part of the question, we treat network N as a graph. Please also list down all possible paths and show which paths are open or closed.

- (c) [5 points] Is h_0 d-separated from h_1 ?
- (d) [5 points] Is i_1 d-separated from o_1 ?
- (e) [5 points] Is i_1 d-separated from o_1 given w_1, w_2 and o_2 ?

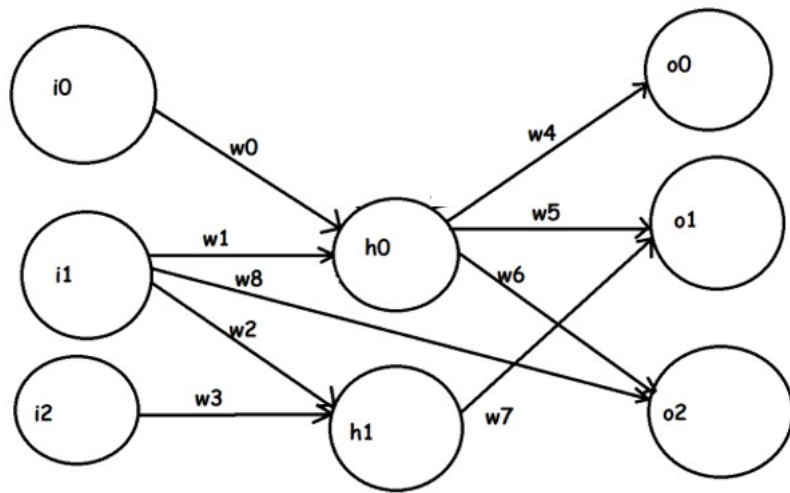


Figure 2: Network N